

# Transfer Learning for Forest Image Classification Using EfficientNet-B2

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**Abstract**—Forest management has become more challenging because of increasing fire risk, drought, and vegetation stress. Manual forest management is time-consuming and difficult to apply across large areas. This study was developed as part of the Sky Innovators Project, which aims to support environmental monitoring and reforestation planning through AI-assisted visual analysis. Three image classification modules were developed and evaluated: forest/non-forest classification, forest density estimation, and fire/no-fire classification. The image datasets were manually reviewed and cleaned before training. All three modules used EfficientNet-B2 with ImageNet pretrained weights. All three models were trained using AdamW and Focal Loss. The forest/non-forest model achieved 99.73% test accuracy and a macro F1-score of 0.9954. The density model achieved 93.04% test accuracy and a macro F1-score of 0.9026. The fire/no-fire model achieved 99.80% test accuracy and a macro F1-score of 0.9922. Since all test sets were drawn from the same main data distributions, these results should be interpreted as internal performance on the prepared datasets rather than as evidence of generalization to external drone, aerial, or satellite imagery. Future work will test the models on independent external datasets.

**Index Terms**—forest/non-forest classification, forest density estimation, fire/no-fire classification, EfficientNet-B2, transfer learning, remote sensing imagery, Focal Loss, class imbalance, environmental monitoring, forest management

## I. INTRODUCTION

### A. Problem Statement and Motivation

Forests are large natural ecosystems primarily covered by trees and vegetation that provide essential ecological, environmental, and climatic functions. According to the Food and Agriculture Organization (FAO), forests cover approximately 31% of the Earth's total land area worldwide [1]. Forests also function as major carbon sinks by absorbing large amounts of atmospheric carbon dioxide ( $\text{CO}_2$ ) through photosynthesis [2]. The United Nations Environment Programme (UNEP) reports that forests absorb approximately 2.6 billion tons of  $\text{CO}_2$  annually, making them highly important for mitigating climate change and reducing greenhouse gas accumulation [2]. Despite their ecological importance, forests are increasingly threatened by wildfires, drought, and deforestation. Wildfires caused approximately 160 million hectares (Mha) of tree cover loss worldwide between 2001 and 2025, accounting for nearly 30% of total global tree cover loss during this period [3]. Drought conditions are among the major factors increasing wildfire frequency and severity worldwide [4]. One of the major challenges for our

environment is deforestation. According to FAO, deforestation resulted in the loss of 10.9 million hectares of forest land in a year [1]. Therefore, strategies for mitigating climate change and maintaining ecological stability should include forest conservation, sustainable management and ecosystem restoration [1]. In conventional forest management practices, manual field surveys and expert knowledge are required [5]. However, these methods are time-consuming, labor-intensive and challenging for large-scale geographic areas [5]. With the increasing availability of remote sensing technology, including UAVs, satellites, large-scale environmental monitoring became more accessible. Extensive forest regions can now be captured rapidly and repeatedly through aerial and satellite imagery [6]. Thus, continuous observation of wildfire risk, vegetation density, drought effects, and land cover changes has become possible [6]. Remote sensing image classification was first used together with traditional machine learning models. However, manual feature extraction techniques are needed for model training. Therefore, machine learning models have difficulty generalizing in different environmental conditions and image acquisition settings [7]. In contrast, deep learning-based approaches can automatically learn hierarchical spatial and textural representations directly from image data, enabling more robust classification performance in remote sensing applications [8]. In particular, convolutional neural network (CNN)-based architectures have demonstrated strong performance in land-cover classification, vegetation analysis, and aerial scene recognition tasks [8]. Architectures such as ResNet, DenseNet, MobileNet, and EfficientNet have therefore become widely adopted in environmental image classification studies due to their ability to efficiently extract complex visual patterns from high-dimensional imagery [8]. Many existing remote sensing studies focus on single-task classification using relatively homogeneous benchmark datasets collected from specific sensors or acquisition conditions. Limited attention is given to iterative dataset cleaning, heterogeneous image sources, and the effect of ambiguous or visually unclear samples on model reliability. In addition, several studies report high classification accuracy without extensively discussing dataset generalization across different regions, seasons, sensors, or acquisition settings. These limitations indicate the need for more application-oriented forest monitoring studies that combine multiple classification tasks with carefully refined datasets and heterogeneous environmental imagery. This study investigates whether EfficientNet-based transfer learning mod-

els can effectively support multiple forest monitoring tasks using heterogeneous RGB image datasets collected from aerial, drone, satellite, and publicly available image sources. Three related classification studies are presented in this study. These are forest/non-forest classification, forest density classification, and fire/no-fire classification. The main contributions of this study are listed below:

- The effects of iterative dataset refinement and manual label correction on the reliability of forest-related image classification tasks are analyzed and discussed.
- A heterogeneous forest/non-forest classification dataset was constructed using aerial, drone, satellite, and publicly available image sources, rather than relying on a single homogeneous dataset.
- All three related forest monitoring models were implemented and analyzed using a unified EfficientNet-B2-based transfer learning framework.
- Challenges associated with environmental imagery, including class overlap, label noise, changing acquisition conditions, were analyzed across all three forest monitoring tasks.
- The limitations of using internal test sets from similar source distributions were explicitly discussed to provide a more transparent interpretation of model generalization performance.

## II. RELATED WORK

Deep learning has become a common approach in remote sensing image classification [8]. Convolutional neural networks can learn spatial and textural patterns directly from image data [8]. Architectures such as ResNet, VGG, DenseNet, MobileNet, and EfficientNet have therefore been used in land-cover classification, scene recognition, and vegetation classification tasks [8].

### A. Forest/Non-Forest Classification

Recent work has investigated deep learning-based forest and vegetation classification using remote sensing imagery. Jain et al. proposed an EfficientNet-B0-based model for forest/non-forest classification using Sentinel-2 satellite imagery [9]. Their model used ImageNet pretrained weights and combined RGB bands with the Normalized Difference Vegetation Index (NDVI) as a fourth input channel [9]. The study reported 98% accuracy and a 98% F1-score, suggesting that EfficientNet-based models can provide strong performance while using fewer parameters than larger CNN architectures such as VGG-16 and ResNet-50 [9]. The authors also emphasized the importance of class balance and diverse non-forest examples for improving generalization [9]. Unlike the multispectral Sentinel-2 imagery used by Jain et al., the present study focuses on heterogeneous RGB imagery collected from multiple aerial, drone, satellite, and publicly available image sources.

### B. Fire/No-Fire Classification

Early wildfire detection is important for reducing ecological damage and improving emergency response efficiency [10].

Recent studies have shown that deep learning-based fire detection systems can support automated real-time wildfire monitoring using surveillance and aerial imagery [10]. CNN-based models are particularly suitable for fire/no-fire classification because they can automatically learn discriminative visual patterns such as flame texture, smoke appearance, and illumination variations from image data [10]. However, wildfire detection remains challenging due to visual similarities between smoke, haze, clouds, sunlight reflections, and bright non-fire regions [10]. Muhammad et al. noted that conventional fire detection systems often struggle in large and complex outdoor environments and may fail to detect fires during early stages [10]. Their study proposed a CNN-based fire detection framework for surveillance applications and achieved 96.8% classification accuracy [10].

### C. Forest Density Classification

Forest density classification has also been studied in drone-based image analysis. CNN-based tasks were suggested by Khidhir and Sehree for forest density classification using UAV images [11]. Images were categorized into three density levels: low, medium and high [11]. VGG-19 and ResNet-50 architectures were employed [11]. ResNet and VGG19 achieved 84.07% and 91.17% classification accuracy, respectively [11]. The authors suggest that larger datasets and extra training are needed for visually similar density classes [11]. This study differs from work conducted by Khidhir and Sehree in terms of two aspects. First, heterogeneous images are used instead of focusing only on UAV imagery. Second, the impact of iterative dataset cleaning on classification reliability is investigated.

### D. Dataset Quality and Generalization Challenges

Land cover classification studies mostly use established remote sensing datasets which include aerial, satellite, or multispectral imagery. The superior performance of deep learning over machine learning is often emphasized in these studies if both models have sufficient labeled data. On the other hand, they rarely discuss dataset quality although it is challenging for these tasks. Publicly available datasets are directly used most of the time in these studies. Manual cleaning, label correction, or the removal of ambiguous samples are often overlooked although model reliability is negatively affected by noisy labels, ambiguous samples and visually unclear images in real-world image classification tasks. For example, a label noise mitigation approach aimed at improving model accuracy was proposed using the Cumulative Spectral Gradient (CSG) method by Gonzalez-Santoyo et al. [12]. Noisy CIFAR-10N and CIFAR-100N datasets were used since they contain labels with different levels of noise [12]. Whereas the model achieved 66% classification accuracy on a dataset containing 40% label noise, the same model achieved 94% classification accuracy on the corresponding clean dataset [12]. Thus, improvements in model precision and classification reliability demonstrate the importance of label correction in image classification.

Similarly, the identification and characterization of label errors were addressed through the Confident Learning approach

proposed by Northcutt et al. [13]. Label issues were identified and removed from ImageNet training data using the Confident Learning method, and the model was retrained from scratch with a ResNet-18 architecture to investigate the effect of removal [13]. The results were compared with a random label removal strategy and showed greater training performance when label issues were removed using the Confident Learning approach [13]. Thus, dataset preparation, label correction, and iterative cleaning were considered critical components of reliable image classification systems. Therefore, this study emphasizes dataset cleaning and manual label refinement as important parts of the classification pipeline.

#### E. Research Gap and Motivation

One limitation in some studies is that high accuracy is reported without clearly stating whether the test data comes from the same source distribution as the training data or from an independent external dataset. This distinction is important because a model may perform well on visually similar test images but perform worse on images collected from different regions, seasons, sensors, or acquisition conditions. Therefore, this study discusses the limitations of using internal test sets for all three modules.

While many land-cover classification studies focus on multi-class problems, forest/non-forest classification is simpler than multi-class land-cover classification and it remains practically useful for identifying forested areas at scale. The density and fire/no-fire modules add finer-grained forest characterization and environmental risk detection, which are relevant for reforestation planning and environmental monitoring. Two widely used benchmark datasets for remote sensing and aerial image classification are EuroSAT and AID [14], [15]. EuroSAT is a satellite-based land-use dataset derived from Sentinel-2 imagery and contains 27,000 images across 10 land-use classes [14]. AID includes 10,000 aerial images across 30 scene categories and is widely used for evaluating aerial scene classification models [15].

This study differs from these benchmark datasets in several ways. First, it addresses three separate classification tasks rather than general multi-class scene recognition: forest/non-forest classification, forest density estimation, and fire/no-fire classification. Second, the datasets were assembled from heterogeneous sources, including drone, aerial, satellite, and other publicly available image sources, rather than from a single sensor or benchmark. Third, the images were manually labeled and refined through iterative cleaning. This study extends previous remote sensing image classification research by focusing more on dataset quality and application-specific classification tasks.

### III. METHODOLOGY

The methodology is divided into three classification modules. Although the modules address different classification tasks, the same EfficientNet-B2 architecture, image resolution, and batch size were used for all of them. The next sections

explain the dataset preparation and training configuration for each module.

#### A. Module 1: Forest/Non-Forest Classification

1) *Dataset Preparation*: The dataset was prepared as a classification dataset with two categories:

- **Forest**: dense forest areas, sparse tree-covered regions, shrubland-like vegetation, and vegetated areas relevant to drought monitoring or reforestation.
- **Non-forest**: urban areas, roads, buildings, agricultural fields, water bodies, bare land, and other non-forested regions.

In total, 34,678 images were manually reviewed and labeled. Images containing tree cover, shrubland-like vegetation, or vegetation relevant to drought monitoring and reforestation were assigned to the forest class.



Fig. 1. Representative examples from the forest and non-forest classes. Left: forest. Right: non-forest.

2) *Iterative Dataset Cleaning*: The forest/non-forest dataset was refined through several experimental iterations. In the early experiments, the model achieved lower accuracy, which motivated a closer review of the dataset and model outputs. During this process, low-quality, ambiguous, visually similar, and incorrectly labeled images were removed or corrected. Samples that did not clearly represent either the forest or non-forest class were also excluded. New higher-quality images were added to improve dataset diversity and quality. After each dataset update, the revised dataset was re-uploaded and the train/validation/test split was recreated.

3) *Data Splitting and Class Imbalance Handling*: The final dataset contained 34,678 images, including 6,980 forest images and 27,698 non-forest images, with an approximate 1:4 class imbalance. To reduce the effect of this imbalance, Focal Loss [16] was used during training, and weighted random sampling was applied at the batch level to oversample the minority forest class. Table I shows the dataset split.

4) *Training Configuration*: The forest/non-forest model used AdamW as the optimizer and Focal Loss as the loss function. The primary checkpoint selection metric was the validation F1-score for the forest class. Training was performed for 30 epochs in two stages using a Kaggle GPU environment. In Stage 1 (epochs 1–10), the EfficientNet-B2 backbone was frozen, and the classifier head was trained with a learning rate of  $1 \times 10^{-3}$ . In Stage 2 (epochs 11–30), the

TABLE I  
FOREST/NON-FOREST CLASSIFICATION DATASET SPLIT

| Set        | Number of Images |
|------------|------------------|
| Training   | 24,274           |
| Validation | 5,202            |
| Test       | 5,202            |
| Total      | 34,678           |

last three feature blocks and the classifier were unfrozen. A differential learning rate strategy was applied:  $1 \times 10^{-5}$  for the unfrozen feature blocks and  $1 \times 10^{-4}$  for the classifier layer. CosineAnnealingLR was used in both stages.

### B. Module 2: Forest Density Classification

1) *Dataset Preparation*: The density dataset contained 6,027 images across three classes: `low_density`, `medium_density`, and `high_density`. These classes correspond to low, medium, and high forest density levels. The dataset split is shown in Table II. The density images were manually reviewed, and unclear or visually ambiguous samples were removed or corrected when necessary.



Fig. 2. Representative examples from the forest density classes. Left: low density. Middle: medium density. Right: high density.

TABLE II  
DENSITY CLASSIFICATION DATASET SPLIT

| Set        | Number of Images |
|------------|------------------|
| Training   | 4,218            |
| Validation | 904              |
| Test       | 905              |
| Total      | 6,027            |

2) *Training Configuration*: The density model used AdamW as the optimizer and Focal Loss with  $\gamma = 2.0$  as the loss function. The primary checkpoint selection metric was the validation macro F1-score. Training was performed in three stages. The best checkpoint was obtained at epoch 35 in Stage 3. It achieved a validation macro F1-score of 0.9126, validation accuracy of 93.25%, and validation loss of 0.1338. The full training configuration is summarized in Table III.

TABLE III  
DENSITY MODEL TRAINING CONFIGURATION

| Component         | Value                         |
|-------------------|-------------------------------|
| Model             | EfficientNet-B2               |
| Image size        | $260 \times 260$              |
| Optimizer         | AdamW                         |
| Loss function     | Focal Loss ( $\gamma = 2.0$ ) |
| Batch size        | 32                            |
| Training schedule | Three-stage fine-tuning       |
| Primary metric    | Validation Macro F1-score     |
| Best epoch        | 35 (Stage 3)                  |
| Best val macro F1 | 0.9126                        |
| Best val accuracy | 93.25%                        |
| Best val loss     | 0.1338                        |

### C. Module 3: Fire/No-Fire Classification

1) *Dataset Preparation*: The fire/no-fire dataset contained 6,601 images across two classes: `fire` and `no_fire`. The dataset split is shown in Table IV. The fire/no-fire images were manually reviewed, and unclear, duplicate, or visually misleading samples were removed when necessary.



Fig. 3. Representative examples from the fire and no-fire classes. Left: fire. Right: no-fire.

TABLE IV  
FIRE/NO-FIRE CLASSIFICATION DATASET SPLIT

| Set        | Number of Images |
|------------|------------------|
| Training   | 4,620            |
| Validation | 990              |
| Test       | 991              |
| Total      | 6,601            |

2) *Training Configuration*: The fire/no-fire model used AdamW as the optimizer and Focal Loss with  $\gamma = 2.0$  as the loss function. The primary checkpoint selection metric was the validation F1-score for the fire class. In Stage 1, only the classifier head was trained, while Stage 2 further fine-tuned the model. The best checkpoint was obtained at epoch 15 in Stage 2. It achieved a validation fire F1-score of 1.0000, validation macro F1-score of 1.0000, validation accuracy of 1.0000, and validation loss of 0.0002. The full training configuration is summarized in Table V.

### D. Shared Components Across All Three Modules

1) *Model Architecture*: All three modules used EfficientNet-B2 [17] with ImageNet pretrained weights [18].

TABLE V  
FIRE/NO-FIRE MODEL TRAINING CONFIGURATION

| Component         | Value                    |
|-------------------|--------------------------|
| Model             | EfficientNet-B2          |
| Image size        | 260 × 260                |
| Optimizer         | AdamW                    |
| Loss function     | Focal Loss (gamma = 2.0) |
| Batch size        | 32                       |
| Primary metric    | Validation Fire F1-score |
| Best epoch        | 15 (Stage 2)             |
| Best val fire F1  | 1.0000                   |
| Best val macro F1 | 1.0000                   |
| Best val accuracy | 1.0000                   |
| Best val loss     | 0.0002                   |

EfficientNet-B2 scales depth, width, and input resolution in a balanced way and provides a good balance between classification performance and training cost. EfficientNet-B2 was chosen for several reasons. First, it uses compound scaling across depth, width, and resolution, which gives good classification performance without needing as many parameters as VGG-16 or ResNet-50. Second, Jain et al. [9] showed that EfficientNet-based models work well on forest classification tasks. Third, the  $260 \times 260$  input resolution fits well with aerial and drone images where spatial detail matters. A comparison with other architectures is planned for future work. Each model was fine-tuned for its task using transfer learning.

2) *Data Augmentation*: Lightweight data augmentation was applied only to the training sets to make the model more robust and reduce the risk of overfitting [19]. The augmentation pipeline included random horizontal flipping ( $p = 0.5$ ), random rotation up to  $\pm 10^\circ$ , and color jitter with brightness, contrast, and saturation factors of 0.15 and a hue factor of 0.03. No augmentation was applied to the validation or test sets. All images were normalized using ImageNet statistics: mean = [0.485, 0.456, 0.406] and standard deviation = [0.229, 0.224, 0.225].

3) *Data Source and Ethics Note*: The datasets were collected from Kaggle and other publicly available sources. During dataset preparation, images from different public sources were combined, manually reviewed, and cleaned through several experimental iterations. Since the dataset was updated multiple times during this process, the exact number of images retained from each original source was not recorded.

The final datasets are therefore reported at the module level rather than at the individual source level. The forest/non-forest module contained 34,678 manually reviewed images, the density module contained 6,027 manually reviewed images, and the fire/no-fire module contained 6,601 manually reviewed images.

The study used aerial, drone, satellite, and land-cover images rather than personal or sensitive data, so no personally identifiable information was used in the experiments.

## IV. EXPERIMENTS AND RESULTS

### A. Module 1: Forest/Non-Forest Classification

During development, multiple experiments were performed to improve both the training pipeline and the dataset quality. In the early experiments, the model achieved approximately 84% accuracy. After improvements in the training setup and dataset preparation, accuracy increased to approximately 92%. Further improvement was achieved by removing ambiguous and low-quality images, correcting mislabeled samples, and adding higher-quality images.

Table VI summarizes the main experimental iterations. In the final experiment, the EfficientNet-B2 model achieved the test performance reported in Table VII. The best checkpoint was obtained at epoch 29, with a validation F1-score of 0.9898 for the forest class.

The confusion matrix for the forest/non-forest model is shown in Table VIII. Among 5,202 test images, only one forest image was misclassified as non-forest, and 13 non-forest images were misclassified as forest.

### B. Module 2: Forest Density Classification

The density model achieved the test performance reported in Table IX. The model performed strongly on *low\_density* and *high\_density* classes, while *medium\_density* was more challenging due to visual overlap with adjacent density levels.

The confusion matrix for the density model is reported in Table X. The most frequent errors involved the *medium\_density* class being confused with *low\_density* or *high\_density*, reflecting the visual ambiguity of intermediate density levels.

### C. Module 3: Fire/No-Fire Classification

The fire/no-fire model achieved the test performance reported in Table XI. The *fire* class achieved an F1-score of 0.9855, while the *no\_fire* class achieved an F1-score of 0.9989, indicating strong internal test performance for distinguishing visible fire patterns from no-fire images.

The confusion matrix for the fire/no-fire model is shown in Table XII. Among 991 test images, 68 *fire* images were correctly classified as *fire*, while one *fire* image was misclassified as *no\_fire*. Similarly, 921 *no\_fire* images were correctly classified, while one *no\_fire* image was misclassified as *fire*.

### D. Cross-Module Comparison

Table XIII provides a summary comparison of the three modules. Forest/non-forest and fire/no-fire models achieved the highest accuracy, which is expected given the clearer visual distinction between their class categories. The density module is more challenging because its categories are finer-grained and visually similar.

## V. DISCUSSION

### A. Forest/Non-Forest Classification

The forest/non-forest model achieved strong performance on the prepared test set, with training and validation curves showing no clear sign of overfitting. During Stage 2, the validation

TABLE VI  
EFFECT OF ITERATIVE DATASET CLEANING ON FOREST/NON-FOREST CLASSIFICATION TEST PERFORMANCE

| Exp.  | Dataset Size | Primary Change                                 | Best Epoch | Test Acc. | Macro F1 | Forest F1 | Non-Forest F1 |
|-------|--------------|------------------------------------------------|------------|-----------|----------|-----------|---------------|
| 1     | ~27,000      | Initial dataset; baseline pipeline             | —          | ~85%      | —        | —         | —             |
| 2     | ~34,750      | Improved pipeline; dataset expanded            | —          | ~92%      | —        | —         | —             |
| 3     | 34,374       | Iterative cleaning; mislabeled images re-moved | 16         | 99.22%    | 0.9866   | 0.9779    | 0.9953        |
| Final | 34,678       | Full cleaning; higher-quality images added     | 29         | 99.73%    | 0.9954   | 0.9925    | 0.9984        |

TABLE VII  
FOREST/NON-FOREST CLASSIFICATION: FINAL TEST RESULTS

| Metric               | Result |
|----------------------|--------|
| Test Loss            | 0.0042 |
| Test Accuracy        | 99.73% |
| Test Macro F1-score  | 0.9954 |
| Forest Precision     | 0.9862 |
| Forest Recall        | 0.9989 |
| Forest F1-score      | 0.9925 |
| Non-Forest Precision | 0.9998 |
| Non-Forest Recall    | 0.9970 |
| Non-Forest F1-score  | 0.9984 |

TABLE VIII  
FOREST/NON-FOREST CLASSIFICATION: CONFUSION MATRIX ON THE FINAL TEST SET

| True \ Predicted | Forest | Non-Forest |
|------------------|--------|------------|
| Forest           | 929    | 1          |
| Non-Forest       | 13     | 4259       |

TABLE IX  
DENSITY CLASSIFICATION: FINAL TEST RESULTS

| Metric                   | Result |
|--------------------------|--------|
| Test Loss                | 0.0927 |
| Test Accuracy            | 93.04% |
| Test Macro F1-score      | 0.9026 |
| Test Weighted F1-score   | 0.9290 |
| low_density Precision    | 0.9066 |
| low_density Recall       | 0.9510 |
| low_density F1-score     | 0.9283 |
| medium_density Precision | 0.8600 |
| medium_density Recall    | 0.7633 |
| medium_density F1-score  | 0.8088 |
| high_density Precision   | 0.9639 |
| high_density Recall      | 0.9776 |
| high_density F1-score    | 0.9707 |

TABLE X  
DENSITY CLASSIFICATION: CONFUSION MATRIX ON THE TEST SET

| True \ Predicted | low | medium | high |
|------------------|-----|--------|------|
| low_density      | 233 | 11     | 1    |
| medium_density   | 23  | 129    | 17   |
| high_density     | 1   | 10     | 480  |

TABLE XI  
FIRE/NO-FIRE CLASSIFICATION: FINAL TEST RESULTS

| Metric                 | Result |
|------------------------|--------|
| Test Loss              | 0.0078 |
| Test Accuracy          | 99.80% |
| Test Macro F1-score    | 0.9922 |
| Test Weighted F1-score | 0.9980 |
| fire Precision         | 0.9855 |
| fire Recall            | 0.9855 |
| fire F1-score          | 0.9855 |
| no_fire Precision      | 0.9989 |
| no_fire Recall         | 0.9989 |
| no_fire F1-score       | 0.9989 |

TABLE XII  
FIRE/NO-FIRE CLASSIFICATION: CONFUSION MATRIX ON THE TEST SET

| True \ Predicted | fire | no_fire |
|------------------|------|---------|
| fire             | 68   | 1       |
| no_fire          | 1    | 921     |

loss remained close to or lower than the training loss: at the best checkpoint, training loss was 0.0045 while validation loss was 0.0034. The validation accuracy was 99.63%, and the test accuracy was 99.73%. This shows that the model performed consistently.

Several factors can explain this strong test performance. First, forest and non-forest images are usually easy to distinguish. Forest images generally include trees and dense vegetation, while non-forest images generally include roads, buildings, and other non-forest areas. Second, iterative cleaning process reduced noisy labels and unclear samples. This improved separation between the classes. Finally, EfficientNet-B2 used ImageNet pretrained weights as an initial base. Fine-tuning on this dataset allowed the model to learn visual patterns related to this task.

TABLE XIII  
CROSS-MODULE PERFORMANCE COMPARISON

| Module            | Classes | Test Acc. | Macro F1 |
|-------------------|---------|-----------|----------|
| Forest/Non-Forest | 2       | 99.73%    | 0.9954   |
| Density           | 3       | 93.04%    | 0.9026   |
| Fire/No-Fire      | 2       | 99.80%    | 0.9922   |



### B. Density Classification

The density model achieved 93.04% test accuracy and a macro F1-score of 0.9026. The *low\_density* and *high\_density* classes had high F1-scores of 0.9283 and 0.9707. The *medium\_density* class had a lower F1-score of 0.8088. The confusion matrix shows that most errors happened in the *medium\_density* class. Some *medium\_density* images were predicted as *low\_density* or *high\_density*. The dataset contained 6,027 images across three classes. One reason for the lower *medium\_density* performance is likely the dataset size. With only 6,027 images across three classes, the model had less data to learn from compared to the forest/non-forest module. The *medium\_density* class is also visually harder to separate from the other two classes. In the next phase of this work, the density dataset can be expanded and the cleaning process can be continued to improve the separation between visually similar density classes.

### C. Fire/No-Fire Classification

The fire/no-fire model achieved 99.80% test accuracy, with strong performance on *fire* (F1: 0.9855) and *no\_fire* (F1: 0.9989). The confusion matrix shows only two misclassifications among 991 test images, suggesting that the model captured visible fire-related patterns well. However, the *fire* class had far fewer test samples than the *no\_fire* class in the test set, with 69 *fire* images compared to 922 *no\_fire* images. It should also be noted that the *fire* class contained only 69 images in the test set. With such a small number of samples, the F1-score of 0.9855 should be read carefully, as a few misclassifications would noticeably change the result. A larger and more varied *fire* test set is needed before drawing strong conclusions about the model's reliability.

### D. Shared Limitations

Across all three modules, the test sets were drawn from the same data distributions as the training and validation sets. No source-wise, geographically independent, or temporally independent split was applied; therefore, visually similar images may exist across the training, validation, and test sets. As a result, the reported results reflect internal test performance, and generalization to external drone, aerial, or satellite imagery remains unverified.

## VI. THREATS TO VALIDITY / LIMITATIONS

- **External validity:** All three test sets were taken from the same data sources as the training and validation sets. This means the results show how well the models perform on familiar data, not on new or unseen imagery. Performance on images from different regions, seasons, sensors, or flight altitudes has not been tested yet, and the current scores should not be treated as a guarantee of real-world performance.
- **Internal validity:** Dataset cleaning for the forest/non-forest module was performed manually, so subjective judgment during labeling and sample removal may have introduced inconsistencies across iterations. The density

and fire/no-fire datasets were also manually reviewed and cleaned but the forest/non-forest dataset had the most extensive iterative cleaning process.

- **Dataset size and class balance:** The density dataset had 6,027 images with three classes. The *medium\_density* class had fewer samples and it was harder for the model to separate it from the *low\_density* and *high\_density* classes. The fire/no-fire dataset was also imbalanced, especially in the test set. The *fire* class had far fewer samples than the *no\_fire* class, so the model may not learn fire images enough.
- **Construct validity:** Accuracy and F1-score are useful metrics, but they do not show everything about the model. For environmental monitoring, the model also needs to work well on difficult images, such as images with different lighting, angles, or unclear forest areas.
- **Conclusion validity:** All experiments were run in a Kaggle GPU environment with a fixed seed of 42. If the same experiments are repeated with different settings, the results may be slightly different.

## VII. CONCLUSION AND FUTURE WORK

This study developed three EfficientNet-B2-based models for forest-related image analysis in the Sky Innovators Project. The forest/non-forest model achieved 99.73% test accuracy and a macro F1-score of 0.9954. The forest density model used 6,027 images across three density classes and achieved 93.04% test accuracy with a macro F1-score of 0.9026. The fire/no-fire model used 6,601 images across two classes and achieved 99.80% test accuracy with a macro F1-score of 0.9922. All datasets were manually reviewed and cleaned before training. These results show that EfficientNet-B2 with transfer learning worked well for the forest classification tasks in this study. However, the results were not the same for each module. The number of classes, dataset size, and similarity between classes affected the performance. The *medium\_density* class had lower performance because it was harder to separate from the other density classes. Also, all test sets came from the same main data sources as the training and validation sets. Therefore, these results show internal test performance, not guaranteed performance on new drone, aerial, or satellite images.

In future work:

- Testing all three models on new external image datasets from different regions, altitudes, seasons, and lighting conditions.
- Expanding the density and fire/no-fire datasets and continuing the cleaning process.
- Comparing EfficientNet-B2 with other architectures such as ResNet, MobileNet, DenseNet, and Vision Transformers.
- GradCAM, which is an explainability method, will be applied to examine model decisions in all three tasks [20] [21]. It will provide visual explanations for regions that contribute most to model predictions. Consequently, it will enhance the interpretability of classification models [20] [21].

## CODE AND DATA AVAILABILITY

The experimental notebooks, datasets, and result reports used in this study are available in a public GitHub repository. This makes it easier to review the work and repeat the experiments.

Repository: <https://github.com/burakefeprojects/paper/>

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